AryaVault: feature list and manual test guide

Windows build, version 0.1.0 (technical preview) · prepared 15 October 2026 from `main` after PR #63 (list cursor, API version 4)

> **Read this first.** AryaVault is an **unaudited, unsigned technical preview**. Test it with **fake data only** and never put your real passwords in a test vault. Do not use it as your only copy of anything. Several things listed here have only been tested by automated tests or with fake components, not by a person on real Windows hardware; this guide is how you close those gaps. Where this guide says *expected*, anything else is a bug worth writing down.

1. Before you start

1.1 Get a build that matches this guide

The newest package in `dist\` was built on 14 October, **before** these changes: list paging by cursor, the new Hello cancel/invalidated messages, and the hidden Print button. To test them you need a fresh build.

> **Known blocker on this PC (15 Oct):** `flutter doctor` reports that the Visual Studio 2026 installation is *incomplete*, so `flutter build windows` fails. Open the **Visual Studio Installer** and use *Repair* (or finish the pending update), then check that `flutter doctor` shows Visual Studio as OK. I did not change your Visual Studio installation.

Then build the portable package (about 5 to 10 minutes):

```
cd "D:\My Work\AryaCode Labs\password-manager"
git pull origin main
powershell -File tools\package\package-windows.ps1
```

This writes `dist\AryaVault-0.1.0-windows-x64-portable.zip` and `SHA256SUMS.txt`. Extract the zip to a new folder, for example `C:\AryaVaultTest\app`. The program is `arya_vault.exe` inside it.

1.2 Always test in a separate profile

AryaVault keeps its vault in `%LOCALAPPDATA%\AryaVault\vault`. To keep tests away from any real data, start the app from PowerShell with `LOCALAPPDATA` pointed at a scratch folder. Each different folder is a separate, empty vault:

```
$env:LOCALAPPDATA = "C:\AryaVaultTest\profile1"
New-Item -ItemType Directory -Force $env:LOCALAPPDATA | Out-Null
& "C:\AryaVaultTest\app\arya_vault.exe"
```

Windows Hello and the Windows credential store do **not** follow this variable: the Hello credential is per Windows user (name `aryavault-quick-unlock-v1`). If you test Hello in two profiles, turning it on in one replaces the credential of the other.

1.3 Fake data and files provided

What	Where / value
Test master password (use for every fake vault)	`CANARY-Orchid-Sandstone-Pilot-6402` (meets the 12-character rule)

Sample CSV to import (3 logins, two share a password, one uses `http://`)	`dist\test-data\sample-import.csv`
Sample Bitwarden JSON (1 login in a folder, 1 note)	`dist\test-data\sample-bitwarden.json`
Pre-built vault with 20,000 fake items (older schema: also tests the upgrade)	`dist\test-data\vault-20k-localappdata` (copy it first; password `CANARY-master-correct-horse-battery-staple-91`)

How to record results: write **P** (pass), **F** (fail) or **S** (skipped) in the Result column and a note for anything odd (what you did, what you saw, a screenshot if the screen is not showing a secret). Screenshots may show only fake data.

2. What is built (feature list)

Status words: **Verified** = automated tests pass and you or I saw it work on Windows. **Partly** = automated tests pass but something specific is not checked on real hardware (listed). **Not built** = does not exist yet.

Area	Features	Status
Create a vault	Master password with live strength meter and a 12-character minimum; confirm field; advanced key-derivation strength (Low / Balanced / High); one-time explanation that there is no password reset.	**Verified**
Recovery key	Shown once after creation; groups with a checksum group; show/hide; no copy button on purpose; *Save as file*; confirmed by typing 3 groups; a wrong answer shows 3 new groups; can be used later to recover the vault. *Print is hidden on Windows by design* (the print spooler would keep a copy of the key on disk).	**Verified**
Lock screen	Unlock with password; wrong-password error and a delay after repeated failures; "Use recovery key" to set a new password; "I don't have my recovery key" information page.	**Verified**
Windows Hello quick unlock	Offered after creating a vault and in Settings; unlock with Hello; password required again after a reboot, after 72 hours, or after 5 failed attempts. New in this build: closing the Hello prompt is silent; a removed or changed Hello credential turns quick unlock off and says so.	**Partly** Turning on and unlocking works. Reboot, cancel, removed credential and enrolment change were only tested with a fake.
Items	Logins, secure notes (Markdown), cards, identities; title, username, password, websites, notes, custom fields (text, hidden, website, date), tags, folders, favorites; masked secrets with reveal that hides itself; copy with automatic clipboard clearing; open website with a confirmation (extra warning for http); unsaved-changes prompt.	**Verified**
Trash and history	Delete to trash with Undo; restore; delete forever; empty trash; purge date; per-field history, compare versions, restore an old version.	**Verified**

Search and lists	Instant search over non-secret fields; All items, Favorites, folders, tags; type filter chips; item counts. Lists load in pages (by cursor): a page of 50 takes about 3 to 5 ms even in a 20,000-item vault (measured in the core, not yet in the on-screen list).	**Partly** On-screen scrolling and memory with 20,000 items not measured.
Password generator	Random passwords (length, character classes, symbol set, leave out look-alikes) and passphrases (words, separator, capitalise, number); entropy estimate; "Use this password" inside the item editor.	**Verified**
Password health	Reused, weak and old passwords; nothing is shown or sent anywhere.	**Verified**
Import and export	Import CSV (generic, Chrome/Edge, Firefox, Safari shapes) and unencrypted Bitwarden JSON with a preview, duplicate skipping and a target folder; AryaVault encrypted export and re-import; plain CSV export behind a typed confirmation. Re-asks the master password first.	**Verified** KeePass (KDBX) import is not built.
Settings	Auto-lock minutes (1 to 60, no "never"), lock on screen lock and on sleep; clipboard clear time (5 to 120 s); reveal auto-hide (1 to 15 s); block screenshots and recording; change master password (with optional key rotation and a new recovery key); view or regenerate recovery key; quick unlock switch; About (versions, licences) and a scrubbed diagnostics report.	**Partly** Lock on minimise exists in the engine but has no switch in Settings.
Windows hardening	Clipboard kept out of Win+V history and cloud clipboard; screen-capture exclusion; lock on workstation lock and on sleep; only one copy of the app can run; vault folder restricted to your Windows user and excluded from Windows Search indexing; crash dialogs and heap dumps suppressed for the app.	**Partly** PrintScreen, Teams/Zoom share and the cloud-clipboard check are not verified. A crash dump of an unlocked app can still be written if your PC has a machine-wide dump policy.
Storage and security core	Argon2id key derivation, XChaCha20-Poly1305 encryption, an encrypted database (nothing readable on disk), safe writes that survive being killed, key rotation, an automatic encrypted backup before a database upgrade.	**Verified** No independent audit and no second human reviewer yet.
Packaging	Portable zip with checksums. The MSIX package **cannot be installed** while unsigned (Windows refuses it), so it is not offered. No code signing yet (Windows SmartScreen will warn).	**Partly**

Not built yet	**Sync and backup to Google Drive / iCloud / a folder; adding a second device; KeePass import; authenticator (TOTP) codes; browser or app autofill; Print; Android, macOS, iOS, Linux; the "erase this vault" option on the forgot-everything page; code signing.**	**Not built**

3. Manual test cases

Do the groups in order the first time: group A creates the vault the later groups use. Use profile `profile1` unless a test says otherwise.

A. Create a vault and the recovery key

#	Steps	Expected	Result
A1	Start the app with an empty profile.	"Welcome to AryaVault" with "Create a new vault". "Add this device to an existing vault" is greyed out ("Available with sync"). Window title reads **AryaVault**.	☐ P ☐ F
A2	Create a vault. Type `short`, then `password1234`.	The meter says Very weak/Weak and the Create button does not proceed. The 12-character message shows.	☐ P ☐ F
A3	Enter the test password in both fields but make the second one differ by a letter. Submit.	"The two passwords don't match." Nothing is created.	☐ P ☐ F
A4	Open *Advanced*. Read the three options. Keep Balanced. Enter the test password twice and create.	Creation takes about a second ("Creating your vault…"). Then the page "There is no password reset".	☐ P ☐ F
A5	Continue to the recovery key. Check there is **no copy button**. Press *Hide key* then *Show key*. Press *Save as file* and save it to a scratch folder.	Key shown in groups with a Checksum group. A hidden placeholder appears when hidden. The system Save dialog opens; the file contains the sheet text and the key. There is **no Print button**.	☐ P ☐ F
A6	Tick "I stored it", continue. On the verify page type three wrong groups.	"That doesn't match…". The page now asks for **different** groups (the core drew new ones).	☐ P ☐ F
A7	Use *Show my key again*, then type the right groups (lower case is fine).	Verification passes and you reach "Unlock faster?" (if Windows Hello is set up on this PC) or the vault.	☐ P ☐ F
A8	Close the app in the middle of onboarding (after creating, before verifying). Start it again.	You cannot reach the vault list without confirming a recovery key: after unlocking you are taken back to the key steps (it may ask for your password to create a new key, because the first can't be shown again).	☐ P ☐ F

B. Lock, unlock, recovery

#	Steps	Expected	Result
B1	Press *Lock now*. On the lock screen type a wrong password.	"That didn't work…"; stays locked. Nothing about which part was wrong.	☐ P ☐ F

B2	Type wrong passwords 6 to 8 times quickly.	After several tries: "Too many attempts. Wait a moment…" (a delay applies). Record how many tries it took.	☐ P ☐ F
B3	Wait, then type the right password and press **Enter** (not the button).	Unlocks.	☐ P ☐ F
B4	Lock. Choose *Use recovery key*. Enter the key you saved (with spaces/dashes, lower case). Choose a new master password.	The vault opens with the **new** password; the **old** password no longer works.	☐ P ☐ F
B5	On the lock screen choose *Use recovery key*, type a key with one character changed.	"That recovery key doesn't look right…" (typo/checksum) or "doesn't open this vault". No crash.	☐ P ☐ F
B6	Choose *I don't have my recovery key*.	The explanation page; nothing is deleted. (The "erase this vault" button is not built yet.)	☐ P ☐ F

C. Items

Create a few items first; later groups use them.

#	Steps	Expected	Result
C1	Use + *New item* → Login. Leave the title empty and save.	"A title is required."	☐ P ☐ F
C2	Create login **Test Bank**: username `alice`, press *Generate a password* and use it, add website `https://bank.example.test`, a tag `money`, a custom Hidden field "PIN" = `4821`, notes. Save.	The item appears in the list and is selected; the details show the password and PIN **masked**.	☐ P ☐ F
C3	Press *Show Password*. Wait without touching.	The password is revealed, then hides itself within the Settings time (default 15 s or less).	☐ P ☐ F
C4	Press *Copy Password*. Paste into Notepad. Wait for the clear time shown in the toast.	"Copied; clears in N s". After N seconds, pasting again gives nothing (or the old non-secret text).	☐ P ☐ F
C5	Press *Open website* on the https item; then add a second website `http://old.example.test` and open that.	A confirmation shows the address; the http one has an extra "unencrypted link" warning. Confirming opens your browser.	☐ P ☐ F
C6	Create a **Secure note** with Markdown (`**bold**`, a list, a link, an image link). Save.	Bold/list render; an image shows "image not loaded" (no network fetch). Link opens only after the confirmation.	☐ P ☐ F
C7	Create a **Card** and an **Identity** with fake numbers.	Card number, security code and PIN are masked; both are saved.	☐ P ☐ F

C8	Edit Test Bank: change the notes, then press Back without saving.	"Discard changes?" Choosing Discard leaves the item unchanged; keeping it returns to the editor.	☐ P ☐ F
C9	Edit the password twice (new values), then open *History* for Password.	Earlier versions are listed with dates; you can *Compare* and *Restore this version*; the restore is itself recorded.	☐ P ☐ F
C10	Mark an item as favorite; add it to a new folder via Manage folders; rename the folder; delete the folder.	Favorites and folder views list it; deleting a folder moves its items to "No folder" and deletes none.	☐ P ☐ F
C11	Move an item to the trash; press *Undo* in the toast. Move it again; open Trash; *Restore* it; move it again; *Delete forever* (confirm).	Each step does what it says; the trash shows a "Deleted for good on …" date; delete-forever asks for confirmation and cannot be undone.	☐ P ☐ F
C12	Trash two items; use *Empty trash*.	A confirmation with the item count; afterwards "The trash is empty."	☐ P ☐ F

D. Search, filters and the list

#	Steps	Expected	Result
D1	Type `test` in Search, then `xyzzy`, then clear it.	Results update as you type, with a count line; "No matching items." for nonsense; clear restores the list.	☐ P ☐ F
D2	Search for text that appears only in a **password** or the hidden PIN (for example `4821`).	No result: secret fields are never searched.	☐ P ☐ F
D3	Use the type filter chips (Login, Note, Card, Identity), including the **Identity** chip.	Each chip is fully visible (not cut off) and filters the list; clearing the chip shows all types again.	☐ P ☐ F
D4	Open *Favorites*, a folder, and a tag in the sidebar. Resize the window narrow, then wide.	Each view shows the right items; the sidebar becomes a menu when narrow; no overflow errors.	☐ P ☐ F

E. Generator and health

#	Steps	Expected	Result
E1	Tools → Password generator. Change the length, switch classes off one at a time, turn all off.	The password and the "about N bits" update; with every class off: "Turn on at least one kind of character."	☐ P ☐ F
E2	Switch to Passphrase; change words, separator, capitalise, add a number. Copy it.	Output follows the options; copying uses the same auto-clear as other secrets.	☐ P ☐ F

E3	Create three logins: two with the same password, one with `password123456`. Open Password health.	The shared password appears under “Reused” (“Shared by 2 items”); the weak one under “Weak”. No password is displayed on this page.	☐ P ☐ F

F. Import and export

#	Steps	Expected	Result
F1	Tools → Import → CSV file → choose `sample-import.csv`.	A preview: 3 items; a note that passwords repeat is fine. Nothing is changed until you press Import.	☐ P ☐ F
F2	Pick a target folder, import. Then import the same file again with “Skip items that already exist” on.	First run: 3 items added. Second run: 3 duplicates skipped, 0 added.	☐ P ☐ F
F3	Import `sample-bitwarden.json` as “Bitwarden export (JSON)”.	1 login (favorite, in a new folder “Imported work”) and 1 note with rendered bold text. A line suggests deleting the import file.	☐ P ☐ F
F4	Try importing a file that is not a vault export (for example a .png renamed .csv).	A clear error or an empty preview; no crash; nothing imported.	☐ P ☐ F
F5	Tools → Export → Encrypted export. Confirm your master password, choose an export password (different), save the `.avex` file.	A Save dialog; “Saved.” Open the file in Notepad: unreadable binary, no item text.	☐ P ☐ F
F6	Import that `.avex` file (type “AryaVault encrypted export”) with a wrong export password, then the right one.	Wrong: “That password doesn’t open this file…”. Right: the preview lists everything as already existing (duplicates).	☐ P ☐ F
F7	Export → Plain CSV. Read the warning; try to continue without typing the phrase; then type `EXPORT`.	The button stays disabled until the phrase is correct. The saved file is readable text with every password — **delete it afterwards**.	☐ P ☐ F

G. Settings

#	Steps	Expected	Result
G1	Settings → Security. Set auto-lock to 1 minute. Do not touch the app for 70 seconds.	The app locks by itself and shows the lock screen. (There is no “never” option.)	☐ P ☐ F
G2	With “When the screen locks” on: press Win+L, sign back in.	The app is locked on return. Turn the setting off and repeat: it stays unlocked.	☐ P ☐ F
G3	With “When the computer sleeps” on: put the PC to sleep (Start → Power → Sleep), wake it.	The app is locked on wake.	☐ P ☐ F
G4	Set clipboard clear to 5 s and reveal hide to 3 s. Copy a password; reveal one.	Clipboard clears after ~5 s; the revealed value hides after ~3 s.	☐ P ☐ F

G5	Settings → Change master password: enter current, a new one; tick *Also rotate keys*. Confirm.	"Master password changed." You are shown a **new recovery key** to save and confirm. After locking, only the new password works and the old recovery key no longer does.	☐ P ☐ F
G6	Change the password again **without** rotating keys.	Works; your **existing** recovery key still works (try it via B4 if you want to be sure).	☐ P ☐ F
G7	Settings → *View recovery key*, then *Make a new recovery key*.	View explains it cannot be shown again; Make-new asks for the master password, warns the old key stops working, and walks through save + verify.	☐ P ☐ F
G8	Settings → About; then Diagnostics → Prepare report.	Shows app, core, API (**4**), format and schema (**3**) versions. The report has no passwords or vault text — read it before saving.	☐ P ☐ F
G9	Settings → Sync & backup.	"Sync is coming soon." Nothing to configure.	☐ P ☐ F

H. Windows Hello (needs Hello set up on this PC) — *the most valuable tests*

#	Steps	Expected	Result
H1	In a fresh profile create a vault; on "Unlock faster?" choose *Turn on* and approve the Hello prompts.	Quick unlock is on (Settings shows the switch on). Several Hello prompts in a row are normal.	☐ P ☐ F
H2	Lock. Press *Unlock with Windows Hello* and approve.	Unlocks without typing the password.	☐ P ☐ F
H3	Lock. Press *Unlock with Windows Hello* and **cancel** the prompt (Esc / close).	**No error message.** You are still on the lock screen; the Hello button is still there and still works; the password works.	☐ P ☐ F
H4	Lock. Press the Hello button and fail the Hello check repeatedly (wrong PIN / finger) 5 times.	After the 5th failed attempt the app requires the master password. Record what each failure shows.	☐ P ☐ F
H5	With quick unlock on, **restart the PC**. Start the app (same profile) and unlock.	Hello is **not** accepted as the first unlock after a reboot: the button may still show, but it tells you to use the master password. After you unlock once with the password, Hello works again.	☐ P ☐ F
H6	Change your Windows Hello PIN or remove and re-add your fingerprint/face (Settings → Accounts → Sign-in options), then lock the app and press the Hello button.	Either it still works (if Windows kept the credential), or you see "Quick unlock was turned off because your device sign-in changed…", the Hello button disappears, and the Settings switch is off. **Write down which happened.**	☐ P ☐ F
H7	Settings → turn quick unlock **off**, then on again (enter the master password).	Off removes the Hello button immediately; on asks for the password and for Hello.	☐ P ☐ F
H8	While turning quick unlock on in Settings, **cancel** the Hello prompt.	No error banner; the switch stays off.	☐ P ☐ F
H9	(Optional, slow) Change the PC date forward by 4 days (or wait 72 h), keep the app locked, try Hello.	The master password is required (72-hour limit). Restore the date afterwards.	☐ P ☐ F

I. Windows protection checks

#	Steps	Expected	Result
I1	Copy a password from the app, then press **Win+V**.	The copied secret is **not** in the clipboard history. (Check Settings → System → Clipboard: cloud sync of clipboard may be on or off; the secret should not sync either way.)	☐ P ☐ F
I2	With "Block screenshots and screen recording" on, take a screenshot with the Snipping Tool (Win+Shift+S) over the app window.	The app window is black or missing in the capture. Turn the setting off: the window is captured normally.	☐ P ☐ F
I3	Press **PrintScreen** with the setting on, paste into Paint.	The window is black/missing. (Not yet verified — please note the result.)	☐ P ☐ F
I4	Share your screen in Teams or Zoom (or record with Xbox Game Bar Win+Alt+R) while the app shows an item.	The app window appears black or hidden to viewers. (Not yet verified.)	☐ P ☐ F
I5	Start the app a second time while it is open (double-click the exe again).	No second window; the existing window comes to the front (or at least nothing new opens).	☐ P ☐ F
I6	In PowerShell run `icacls "$env:LOCALAPPDATA\AryaVault\vault"` with the same `LOCALAPPDATA` as the test.	Only `<PC>\<you>:(OI)(CI)(F)` is listed (no Administrators, SYSTEM, Users, Everyone).	☐ P ☐ F
I7	Run `attrib "$env:LOCALAPPDATA\AryaVault\vault\vault.db"`. Then open `vault.db` in Notepad (with the app locked or closed).	The **I** (not content indexed) flag is shown. Notepad shows unreadable data and does **not** start with "SQLite format 3"; none of your item titles or passwords appear when you search the text.	☐ P ☐ F
I8	Lock the app. Search the whole profile folder for a unique title you used (PowerShell: `Select-String -Path "$env:LOCALAPPDATA\*" -Pattern "Test Bank" -Recurse`).	No match in any file.	☐ P ☐ F
I9	While unlocked, kill the app from Task Manager (End task). Start it again.	The vault opens normally after unlocking; nothing is lost or half-written.	☐ P ☐ F

J. The 20,000-item vault (performance and upgrade)

1. Copy the folder `dist\test-data\vault-20k-localappdata` to `C:\AryaVaultTest\big`.
2. Start the app with `$env:LOCALAPPDATA = "C:\AryaVaultTest\big"`.

#	Steps	Expected	Result
J1	Note the time from double-click until the lock screen is visible. Unlock with `CANARY-master-correct-horse-battery-staple-91`.	Lock screen within about 2 s (earlier measurement: window in 1.6 s warm). Unlock takes about a second; the vault then upgrades silently.	☐ P ☐ F
J2	Look in `C:\AryaVaultTest\big\AryaVault\vault`.	A new file `vault.db.bak-v2-…` exists: the encrypted backup made before upgrading the database from schema 2 to 3. It is encrypted and kept for 14 days.	☐ P ☐ F
J3	Watch the list. Scroll with the mouse wheel and by dragging the scrollbar to the bottom. Keep scrolling past where it loads more.	Smooth scrolling; more items load as you reach the end; no gaps, no duplicate rows at page boundaries. Note any stutter.	☐ P ☐ F
J4	Open Task Manager → Details → `arya_vault.exe`. Note the memory while unlocked, after scrolling to the end, and after locking.	Record the numbers. (Lock screen with this vault: about 205 to 219 MB. The unlocked figure is not known yet.)	☐ P ☐ F
J5	Search for `title-12345`, then `canary`.	Results appear almost instantly (core: 30 to 60 ms).	☐ P ☐ F
J6	Edit an item near the end of the list, save, and go back to the list.	The list reloads from the top; the edited item is first; no crash or repeated rows.	☐ P ☐ F

K. Package and install

#	Steps	Expected	Result
K1	In PowerShell run `Get-FileHash dist\AryaVault-0.1.0-windows-x64-portable.zip -Algorithm SHA256` and compare with `dist\SHA256SUMS.txt`.	The hashes are identical.	☐ P ☐ F
K2	Extract the zip to a new folder and start `arya_vault.exe` by double-click (not from PowerShell). Note that this uses your real `%LOCALAPPDATA%`: only do it if you are happy to have a throwaway vault there, or skip this test.	Windows SmartScreen may warn (the app is unsigned): that is expected. The app starts.	☐ P ☐ F
K3	Delete `arya_vault_ffi.dll` from a *copy* of the extracted folder and start that copy.	"AryaVault can't start" with a reinstall message (no crash, no blank window).	☐ P ☐ F

4. Known limits you will meet (not bugs)

- Sync, adding another device, KeePass import, authenticator codes, autofill and Print do not exist yet.
- The MSIX file in `dist` cannot be installed on Windows because it is unsigned. Use the zip.
- Item history shows "edited on another device" entries only after sync exists, so you will only see this device's versions.

- The Hello button goes away only when the core turns quick unlock off (invalidated credential, rotation, password-only policy limits); a one-off failure just shows a message.
- If your PC has a machine-wide crash-dump policy, a crash while the vault is *unlocked* could write a dump file containing keys. Do not test with real data.
- No independent security audit has been done; no second human has reviewed the cryptography and API code.

5. When something fails

Write down: the test number, what you did, what you expected, what happened, and the time. Do **not** paste real secrets. For crashes or odd states you can save *Settings → About → Diagnostics* (a scrubbed report) and send me that together with the test number. If the app will not start, note whether the profile folder was fresh.

Summary sheet

Group	Tests	Passed	Failed	Notes
A Create and recovery key	8			
B Lock, unlock, recovery	6			
C Items	12			
D Search and lists	4			
E Generator and health	3			
F Import and export	7			
G Settings	9			
H Windows Hello	9			
I Windows protection	9			
J 20,000-item vault	6			
K Package and install	3			